

KPC Secure Operational Data Pipeline

Board Pitch | Week 11 Strategic Business Case

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DECISION ASK

GO

Approve a stage-gated implementation,
starting with an 8-week pilot.

Business case uses planning assumptions that must be validated with representative KPC data.

The operational problem

Trusted analysis is slowed by data clean-up, privacy handling and release risk.

1. Manual rework

Operational records can arrive incomplete, duplicated or outside expected ranges. Analysts spend time fixing issues before the data can be trusted.

2. Sensitive data

Staff IDs, truck plates and location details must be protected before wider analytics use. A missed field can create avoidable compliance exposure.

3. Deployment risk

A broken analytics release can reduce trust or disrupt reporting. Changes need testing, health checks and a safe way to return to the last working version.

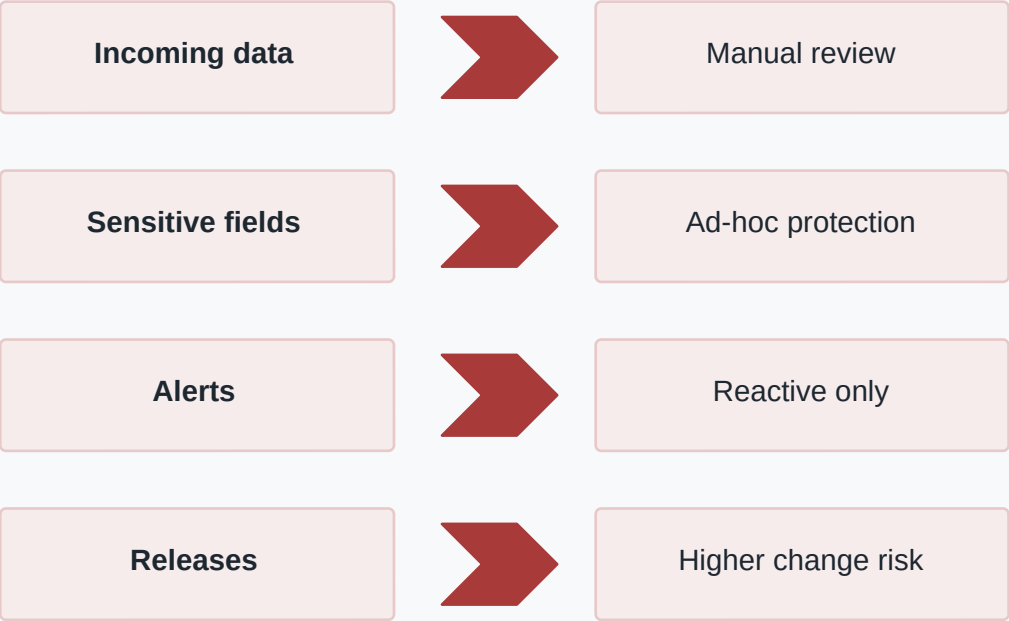
Business consequence

More analyst hours spent on clean-up | slower decisions | higher privacy and operational risk

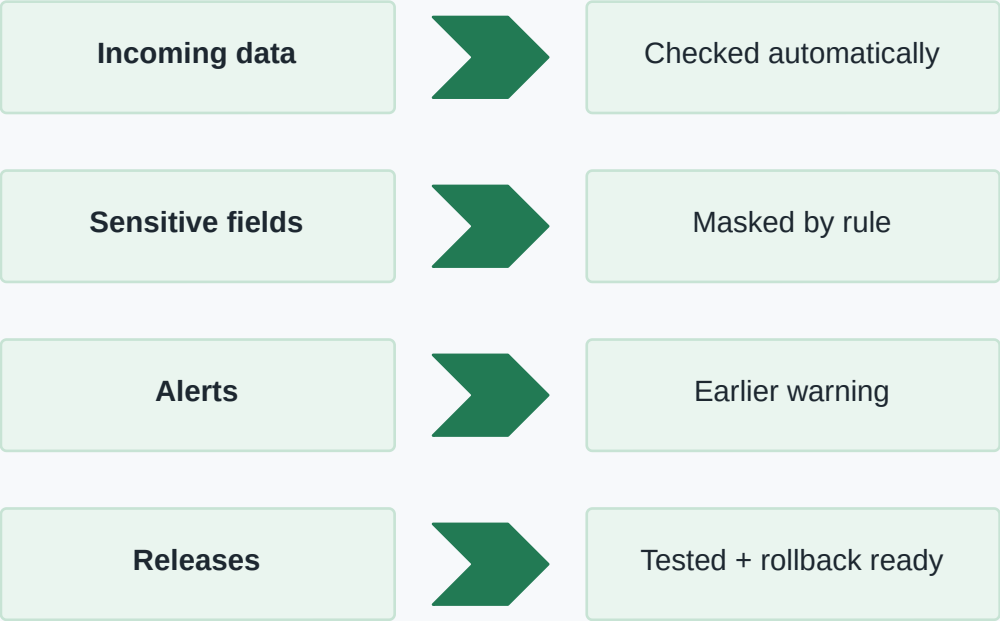
What changes

One controlled path from incoming records to decision-ready information.

TODAY



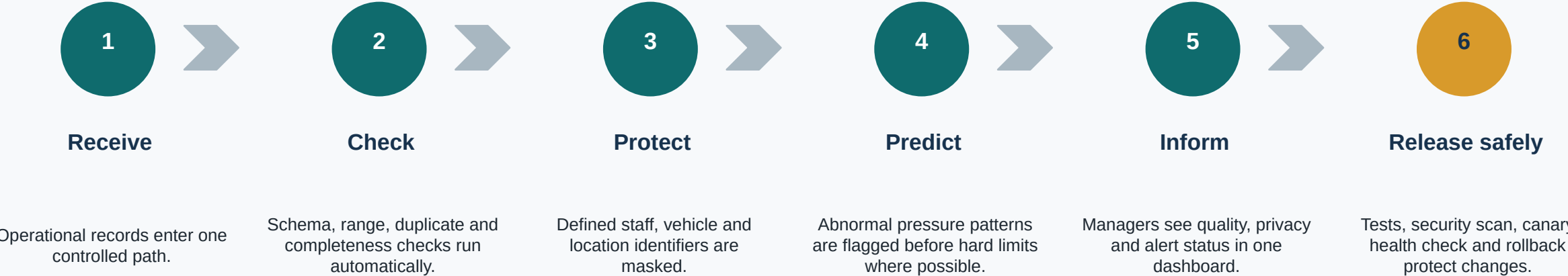
WITH THE PIPELINE



Result: less avoidable rework, safer data use and faster confidence in operational decisions.

How the solution works

Simple controls at each step - no single point of trust.



What is already built in the prototype

Automated checks • sensitive-field masking • predictive alerts • executive dashboard • unit tests • security scan
• Docker build • canary health check • rollback path

Prototype runs on synthetic Phase 1 data; production value must be proven with representative KPC data.

How success will be measured

Board-level gates tie the technical solution to operational outcomes.

$\geq 95\%$

accepted-record target

100%

defined fields masked

$< 10\%$

noisy-alert target

$\geq 99.5\%$

service-uptime target

Operational outcomes to validate during the pilot

- How many analyst hours are actually spent reviewing bad records today?
- How much review time disappears after automated checks?
- Do managers receive earlier, useful alerts instead of more noise?
- Can the team release changes safely without interrupting the last working version?

Only scale if both the control metrics and the labour-saving evidence hold.

Financial case

Planning estimate: attractive return, but only if pilot assumptions are confirmed.

KES 5.9M

net recurring annual savings

171%

Year-1 ROI

3.7 months

payback period

Savings math

$25,000 \text{ records/week} \times 5\% \text{ manual review} \times 4 \text{ minutes} / 60 \times \text{KES } 1,500/\text{hour} \times 52 = \text{KES } 6.50\text{M}$ gross annual labour cost avoided.

Less KES 0.60M annual support = KES 5.90M net recurring savings.

Investment & return

One-time implementation: KES 1.80M. Year-1 cost = KES 2.40M. $\text{ROI} = (6.50 - 2.40) / 2.40 = 171\%$. $\text{Payback} = 1.80 / (5.90 / 12) = 3.7 \text{ months}$.

Assumptions: project dashboard default volume, review time and labour cost; 5% review rate derived from the 95% quality target. Investment/support costs are planning estimates.

Top risks and how we control them

The recommendation is a controlled GO, not a blind full rollout.

Risk 1 - the savings case is wrong

Mitigation

- Run an 8-week pilot on representative operational data.
- Measure actual record volume, manual review time and labour cost.
- Recalculate ROI before scaling.
- Stop if the value case does not hold.

Risk 2 - automation misses a critical issue

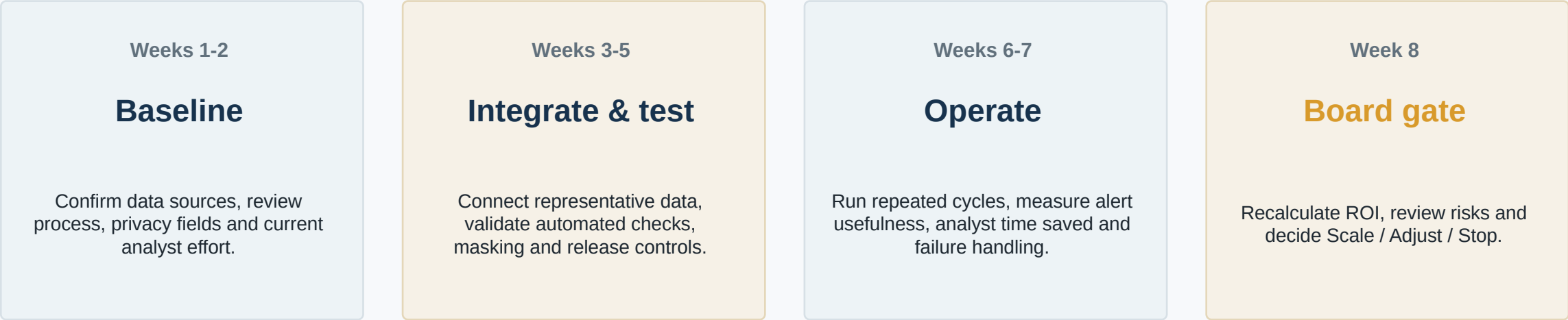
Mitigation

- Keep a human override and visible exception queue.
- Verify 100% masking of defined sensitive fields.
- Track false alerts and tune thresholds.
- Restrict access and retain automatic rollback.

Governance rule: no scale-up decision until the pilot proves both control quality and financial value.

8-week pilot plan

A small, measurable test before any production-scale commitment.



Pilot exit criteria

>=95% accepted records • 100% masking of defined identifiers • <10% noisy alerts • reliable release health checks • validated labour-saving evidence

Recommendation

GO - approve a stage-gated implementation, beginning with the 8-week pilot.

GO

Why now

The prototype already demonstrates the core controls. The next uncertainty is not whether the code can run - it is whether the value case holds on representative KPC data.

Board approvals requested

1

Executive sponsor

One accountable sponsor to remove blockers and own the pilot decision.

2

Representative data access

A controlled dataset that reflects real operational patterns and privacy requirements.

3

Staged budget ceiling: KES 1.80M

Release funds by stage; do not scale if the success gates fail.

Decision at Week 8: SCALE | ADJUST | STOP